

A LINEAR-TIME ALGORITHM FOR THE QUADRATIC FORM ON THE GAP POSET OF A NUMERICAL SEMIGROUP, WITH GENERALISATION TO MULTIPLE GENERATORS

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ABSTRACT. Let a, b be coprime positive integers with $a < b$, and let G be the gap set of the numerical semigroup $\langle a, b \rangle$, ordered as $g_0 < g_1 < \dots < g_{N-1}$ where $N = \frac{(a-1)(b-1)}{2}$. Huang [3] introduced a quadratic form

$$Q(\mathbf{n}) = \sum_{i,j \in G} K(g_j - g_i) n_i n_j,$$

where K is an inclusion–exclusion kernel, and showed that Q recovers the dinv statistic from algebraic combinatorics. The natural evaluation of Q requires $O(N^2)$ arithmetic operations.

We observe that the kernel $K(d)$ is constant on four arithmetic intervals and vanishes identically for $d < 0$ and $d \geq a + b$. This arithmetic structure allows the double sum to be condensed into two sliding-window sums, yielding an $O(N)$ algorithm and reducing complexity from $O(a^2b^2)$ to $O(ab)$.

We then generalise to numerical semigroups $\langle p_1, \dots, p_n \rangle$ with n generators. The generalised kernel $K^{(n)}$ is defined by inclusion–exclusion over all 2^n subsets of generators. We show that the number of active windows $w(n)$ grows strictly slower than 2^n , due to cancellations arising from redundant subset sums of mixed parity. The resulting algorithm runs in $O(w(n) \cdot N)$. While $w(n)$ grows with n , it grows strictly slower than 2^n : the ratio $w(n)/(2^n - 1)$ decreases in direct analogy to the Mertens sieve product $\prod (1 - 1/p_i)$ (Theorem 8.4); the gap set size N does not increase when generators are added. Absolute complexity therefore depends on generator density — for small dense generators N collapses, for sparse ones N dominates. The asymptotic behaviour is verified numerically across eight qualitatively distinct generator families up to $n = 300$.

1. INTRODUCTION

Let $a, b \in \mathbb{N}$ with $\gcd(a, b) = 1$ and $a < b$. The *numerical semigroup* generated by a and b is

$$\langle a, b \rangle = \{xa + yb : x, y \in \mathbb{Z}_{\geq 0}\}.$$

By the Sylvester–Frobenius theorem, the complement $G = \mathbb{N} \setminus \langle a, b \rangle$ is finite with $|G| = N := \frac{(a-1)(b-1)}{2}$. We write $G = \{g_0, g_1, \dots, g_{N-1}\}$ with $g_0 < g_1 < \dots < g_{N-1}$.

In the recent preprint [3], Huang defines a quadratic form on $\mathbb{R}^G$ in terms of the kernel function

$$K(d) := \mathbf{1}_{d \geq 0} - \mathbf{1}_{d \geq a} - \mathbf{1}_{d \geq b} + \mathbf{1}_{d \geq a+b}, \quad d \in \mathbb{Z},$$

and proves that this form recovers the dinv statistic for rational Dyck paths. A Lean 4 formalization of this result was carried out by AxiomProver [1].

Both works address the *correctness* of the identity $Q = \text{dinv}$. The present note addresses the orthogonal question of computational efficiency and its generalisation to n generators.

Relation to the Frobenius problem. The generalisation to n generators does *not* address the Frobenius problem for $n \geq 3$, which asks for a closed formula for the largest integer not in $\langle p_1, \dots, p_n \rangle$ and remains open for $n \geq 3$ [4].

The present work operates under a fundamentally different question. The kernel $K^{(n)}$ is defined solely by inclusion–exclusion over the generator set, and its interval structure follows

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directly from this definition, independently of the Frobenius number or the genus of the semigroup. The sliding-window algorithm takes the sorted gap set G as input — it does not compute G from generators. The computational barrier of the Frobenius problem therefore does not apply here.

This observation is itself enabled by Huang's formulation: it is precisely the inclusion–exclusion structure of K that makes the arithmetic of $K^{(n)}$ transparent and its generalisation direct. Huang's definition, designed for a geometric purpose, incidentally exposes a structure that extends well beyond its original scope.

Main results.

Theorem 1.1 (Linear evaluation, two generators). *For any $\mathbf{n} \in \mathbb{R}^G$,*

$$Q(\mathbf{n}) = \sum_{k=0}^{N-1} n_k (W_k^+ - W_k^-),$$

where

$$W_k^+ = \sum_{j: g_k \leq g_j < g_k + a} n_j,$$

$$W_k^- = \sum_{j: g_k + b \leq g_j < g_k + a + b} n_j.$$

Both families can be computed in $O(N)$ total operations via two sliding windows.

Theorem 1.2 (Generalised sliding-window evaluation). *Let $\mathbf{p} = (p_1, \dots, p_n)$ with $\gcd(p_1, \dots, p_n) = 1$, and let G be the gap set of $\langle p_1, \dots, p_n \rangle$. Define*

$$K^{(n)}(d) = \sum_{S \subseteq \mathbf{p}} (-1)^{|S|} \mathbf{1}_{d \geq \sigma(S)}, \quad \sigma(S) = \sum_{p \in S} p.$$

Let $w(n)$ denote the number of intervals on which $K^{(n)}$ is nonzero. Then $Q(\mathbf{n})$ is computable in time $O(T_{\text{prep}} + w(n) \cdot N)$, where $T_{\text{prep}} = O(n \cdot \sigma_n)$ via the incremental algorithm (Theorem 1.3).

Theorem 1.3 (Incremental window computation). *The active window set for n generators can be computed incrementally from the active window set for $n - 1$ generators in $O(\sigma_{n-1})$ time, where $\sigma_{n-1} = \sum_{i=1}^{n-1} p_i$. The total preprocessing cost for all n generators is $O(n \cdot \sigma_n)$, compared to $O(2^n)$ for direct enumeration. For prime generators ($p_i = i$ -th prime), $\sigma_n = \Theta(n^2 \log n)$ by the prime number theorem, giving $O(n^3 \log n)$.*

2. ARITHMETIC STRUCTURE OF THE KERNEL

2.1. Two generators.

Lemma 2.1 (Interval decomposition of K). *For all $d \in \mathbb{Z}$:*

$$K(d) = \begin{cases} 0 & \text{if } d < 0, \\ +1 & \text{if } 0 \leq d < a, \\ 0 & \text{if } a \leq d < b, \\ -1 & \text{if } b \leq d < a + b, \\ 0 & \text{if } d \geq a + b. \end{cases}$$

Proof. By direct computation from $K(d) = \mathbf{1}_{d \geq 0} - \mathbf{1}_{d \geq a} - \mathbf{1}_{d \geq b} + \mathbf{1}_{d \geq a+b}$, using $a < b$:

d	$\mathbf{1}_{d \geq 0}$	$-\mathbf{1}_{d \geq a}$	$-\mathbf{1}_{d \geq b}$	$+\mathbf{1}_{d \geq a+b}$	$K(d)$
$d < 0$	0	0	0	0	0
$0 \leq d < a$	1	0	0	0	+1
$a \leq d < b$	1	1	0	0	0
$b \leq d < a+b$	1	1	1	0	-1
$d \geq a+b$	1	1	1	1	0

The five cases are exhaustive and mutually exclusive. $\square$

3. REDUCTION OF THE DOUBLE SUM

Lemma 3.1 (Support restriction). *Every term in*

$$(1) \quad Q(\mathbf{n}) = \sum_{k=0}^{N-1} \sum_{\ell=0}^{N-1} K(g_\ell - g_k) n_k n_\ell$$

with $g_\ell - g_k < 0$ or $g_\ell - g_k \geq a+b$ contributes zero.

Proof. Immediate from $K(d) = 0$ for $d < 0$ and $d \geq a+b$ (Lemma 2.1). $\square$

Lemma 3.2 (Interval separation). *For fixed k ,*

$$\sum_{\ell: 0 \leq g_\ell - g_k < a+b} K(g_\ell - g_k) n_\ell = W_k^+ - W_k^-.$$

Proof. From Lemma 2.1: $K = +1$ on $[0, a)$ and $K = -1$ on $[b, a+b)$, zero otherwise. $\square$

Proof of Theorem 1.1. Combine Lemmas 3.1 and 3.2. $\square$

4. LINEAR-TIME ALGORITHM

Theorem 4.1 (Complexity, two generators). $(W_k^+ - W_k^-)_{k=0}^{N-1}$, and hence $Q(\mathbf{n})$, can be computed in $O(N)$ time and $O(N)$ space.

Proof. Maintain four pointers $\text{lo}^+, \text{hi}^+, \text{lo}^-, \text{hi}^-$ into sorted G , initialized to 0, with running sums $W^+ = 0$ and $W^- = 0$. For each $k = 0, 1, \dots, N-1$:

- (i) advance hi^+ while $g_{\text{hi}^+} < g_k + a$, adding n_{hi^+} to W^+ ;
- (ii) retract lo^+ while $g_{\text{lo}^+} < g_k$, subtracting n_{lo^+} from W^+ ;
- (iii) advance hi^- while $g_{\text{hi}^-} < g_k + a + b$, adding n_{hi^-} to W^- ;
- (iv) retract lo^- while $g_{\text{lo}^-} < g_k + b$, subtracting n_{lo^-} from W^- ;
- (v) accumulate $n_k \cdot (W^+ - W^-)$.

Each pointer traverses G at most once: total $\leq 4N$ updates. $\square$

Corollary 4.2. *In terms of the generators, $N = \frac{(a-1)(b-1)}{2} = O(ab)$, so the algorithm runs in $O(ab)$ time versus $O(a^2b^2)$ naive.*

5. INCREMENTAL COMPUTATION OF ACTIVE WINDOWS

The active window set $\mathcal{W}$ for n generators requires knowing all subset sums of $\{p_1, \dots, p_n\}$. Direct enumeration costs $O(2^n)$. We show this can be reduced to $O(n \cdot \sigma_n)$ where $\sigma_n = \sum_{i=1}^n p_i$.

Definition 5.1 (Parity balance). For each value $s \in \mathbb{N}$, define the *parity balance*

$$\pi_n(s) = \sum_{\substack{S \subseteq \{p_1, \dots, p_n\} \\ \sigma(S) = s}} (-1)^{|S|}.$$

Then $K^{(n)}(d) = \sum_{s \leq d} \pi_n(s)$ (prefix sum of π_n).

Lemma 5.2 (Incremental update). *Adding generator p_{n+1} updates the parity balance as follows:*

$$\pi_{n+1}(s) = \pi_n(s) - \pi_n(s - p_{n+1}),$$

where by convention $\pi_n(s) = 0$ for $s \notin \{0, \dots, \sigma_n\}$, and the singleton $\{p_{n+1}\}$ is captured automatically through $\pi_n(0) = 1$ (the contribution of the empty subset).

Proof. Subsets of $\{p_1, \dots, p_{n+1}\}$ with sum s are either: subsets $S \subseteq \{p_1, \dots, p_n\}$ with $\sigma(S) = s$, contributing $(-1)^{|S|}$ each — their sum is $\pi_n(s)$; or subsets of the form $S \cup \{p_{n+1}\}$ where $S \subseteq \{p_1, \dots, p_n\}$ with $\sigma(S) = s - p_{n+1}$, contributing $(-1)^{|S|+1} = -(-1)^{|S|}$ each — their sum is $-\pi_n(s - p_{n+1})$. The singleton $\{p_{n+1}\}$ corresponds to $S = \emptyset$ in the second class at $s = p_{n+1}$, contributing $-\pi_n(0) = -1$, as expected. $\square$

Proof of Theorem 1.3. Each update step processes at most $|\text{supp}(\pi_n)| \leq \sigma_n$ values. Summing over n steps gives total cost $\sum_{k=1}^n \sigma_k = O(n \cdot \sigma_n) = O(n^3 \log n)$ for prime generators. $\square$

Remark 5.3 (Benchmark). The incremental algorithm computes active windows for $n = 2, \dots, 30$ without the exponential cost of direct subset enumeration. As a cross-check, direct $O(2^n)$ enumeration was run independently (Section 7.1) and reached $n = 28$, confirming agreement with the incremental results at $n = 27$ and $n = 28$ on both experimental platforms.

Remark 5.4. The original implementation of Q in [3] and its Lean 4 formalization [1] evaluate the double sum (1) directly, requiring N^2 operations per input vector; the sliding-window algorithm requires $4N$. The ratio

$$\frac{N^2}{4N} = \frac{N}{4} = \frac{(a-1)(b-1)}{8}$$

grows linearly with N and is therefore unbounded. For $(a, b) = (97, 101)$ with $N = 4801$ this gives 23 049 601 versus 19 204 operations, a ratio of approximately 1 200; systematic numerical studies on the original formulation become prohibitive at this scale.

Regarding numerical precision: the observed discrepancy of 1.3×10^{-9} at $N = 4801$ is not algorithmic disagreement but floating-point accumulation. The naive $O(N^2)$ sum accumulates rounding errors at rate $\sim N^2 \varepsilon |Q|$, while the sliding-window algorithm accumulates at rate $\sim N \varepsilon |Q|$, where $\varepsilon \approx 2.2 \times 10^{-16}$ is machine epsilon. Both algorithms agree with each other to within 2×10^{-10} at $N = 4801$ (Table 1; prefix and linear columns). The theoretical threshold at which naive accumulation reaches $|\text{error}| = 1$ is $N \approx 165\,000$, corresponding to generators $(a, b) \approx (406, 408)$; for the sliding-window algorithm this threshold is $N \approx 10^8$. All results in Table 1 are within full double-precision accuracy.

6. GENERALISATION TO n GENERATORS

6.1. Generalised kernel and active windows.

Definition 6.1 (Generalised kernel). Let $\mathbf{p} = (p_1, \dots, p_n)$ with $\gcd(p_1, \dots, p_n) = 1$.

$$K^{(n)}(d) = \sum_{S \subseteq \{p_1, \dots, p_n\}} (-1)^{|S|} \mathbf{1}_{d \geq \sigma(S)}, \quad \sigma(\emptyset) = 0.$$

Lemma 6.2 (Vanishing outside $[0, \sum p_i]$). $K^{(n)}(d) = 0$ for $d < 0$ and for $d \geq \sum_{i=1}^n p_i$.

Proof. For $d < 0$: every indicator is zero. For $d \geq \sum p_i$: all indicators equal 1, and $\sum_{S \subseteq \mathbf{p}} (-1)^{|S|} = (1 - 1)^n = 0$ by the binomial theorem. $\square$

Definition 6.3 (Active window). An interval $[s_k, s_{k+1})$ between consecutive breakpoints of $K^{(n)}$ is *active* if $K^{(n)}$ is nonzero on it. Let $w(n) = w(\mathbf{p})$ denote the number of active windows.

6.2. Redundant subset sums and parity cancellation.

Definition 6.4 (Redundant subset sum). A value $s \in \mathbb{N}$ is a *redundant subset sum* if $s = \sigma(S_1) = \sigma(S_2)$ for two distinct subsets $S_1 \neq S_2$. It is *cancelling* if $|S_1| \not\equiv |S_2| \pmod{2}$.

Lemma 6.5 (Cancellation mechanism). *Let s be a cancelling redundant subset sum. Then subsets of opposite parity contribute opposite signs to $K^{(n)}(s)$, reducing $|K^{(n)}(s)|$ relative to the non-redundant case and potentially driving $K^{(n)}(s) = 0$, thereby eliminating a breakpoint from the active set.*

Proof. Direct from Definition 6.1: each subset S contributes $(-1)^{|S|}$ at every point $d \geq \sigma(S)$. Two subsets with equal sum and opposite parities contribute $+1$ and -1 at the same breakpoint. $\square$

Lemma 6.6 (Linear upper bound on $w(n)$). *For any generator sequence $\mathbf{p} = (p_1, \dots, p_n)$ with $\gcd(p_1, \dots, p_n) = 1$,*

$$w(n) \leq \sigma_n = \sum_{i=1}^n p_i.$$

Proof. Every breakpoint of $K^{(n)}$ is a subset sum $\sigma(S)$ for some $S \subseteq \mathbf{p}$, and all such sums lie in $[0, \sigma_n] \cap \mathbb{Z}$, which contains $\sigma_n + 1$ integers. The active windows are intervals $[s_k, s_{k+1})$ between consecutive breakpoints with $K^{(n)}$ nonzero; their number is at most the number of breakpoints minus one, which is at most σ_n . $\square$

Corollary 6.7 (Asymptotic strict inequality). *For any sequence with $M(n) = \prod(1 - 1/p_i) \rightarrow 0$, the ratio $w(n)/(2^n - 1)$ tends to 0 asymptotically faster than $M(n)$. In particular $w(n) < 2^n - 1$ for all sufficiently large n .*

Proof. Immediate from Theorem 8.4. $\square$

Remark 6.8 (Empirical strict inequality at finite n). Numerical data (Table 3; eight sequence families to $n = 300$ in `multi_sequence_results.txt`) confirm $w(n) < 2^n - 1$ and $w(n)/(2^n - 1)$ monotonically decreasing for every $n \geq 3$ tested, with the redundancy rate (Lemma 6.5) growing from 0% at $n = 2$ to 100% from $n = 21$ onward when generators are the first n primes. A combinatorial proof of strict inequality at every finite $n \geq 3$ for arbitrary coprime sequences is left as an open problem.

6.3. Algorithm and complexity.

Proof of Theorem 1.2. Compute the active window set $\mathcal{W}$ via the incremental algorithm of Theorem 1.3 in time $T_{\text{prep}} = O(n \cdot \sigma_n)$. For each $(lo, hi, c) \in \mathcal{W}$, run one sliding-window pass over sorted G in $O(N)$ time. Total cost is therefore $O(T_{\text{prep}} + w(n) \cdot N)$. $\square$

Remark 6.9 (Empirical cost reduction). Adding generator p_{n+1} produces a new gap set G' with $|G'| \leq |G|$ (adding generators can only enlarge the semigroup), and increases the window count by $\Delta w(n+1) := w(n+1) - w(n)$. Empirically (see Section 10, item (vii)) $\Delta w(n) \approx p_n$ for large n , while $w(n) \approx \sigma_n$, so the relative increment $\Delta w(n)/w(n)$ behaves as $1/n$. Both factors of the cost $w(n) \cdot N$ thus tend to diminish per added generator, in informal analogy to the prime sieve; a formal asymptotic statement requires Open Problem (vii).

7. NUMERICAL VERIFICATION

7.1. Experimental setup. Numerical experiments were carried out in Python 3.14 (with SymPy 1.14 and NumPy) on two independent platforms: a workstation with an AMD Ryzen 9 5900X processor (12 cores, 24 threads) and 64 GB of memory; and a laptop with an Intel Core i7-13620H processor (6P+4E cores, 16 threads) and 16 GB of memory. Both machines run Windows 11 Pro with identical Python and library versions. Floating-point computation uses IEEE 754 double precision (machine epsilon $\varepsilon \approx 2.2 \times 10^{-16}$). Reference scripts: `verify_linear_Q.py`, `verify_v2.py`, `incremental_windows.py`.

Throughout this section, an *operation* denotes one execution of the inner-loop body of the corresponding procedure: in the naive $O(N^2)$ algorithm, one term $K(g_\ell - g_k) n_k n_\ell$ of the double sum (1); in the sliding-window $O(N)$ algorithm, one pointer advance with the associated update of the running window sum. Operation counts are deterministic functions of the input pair (a, b) alone, not wall-clock measurements.

Computations using direct $O(2^n)$ subset enumeration — employed as a cross-check against the incremental algorithm at $n = 27$ and $n = 28$ — use multi-threaded execution; all other reported quantities are produced single-threaded and are bit-identical across both platforms.

7.2. Two generators. Three implementations — naive $O(N^2)$, prefix-sum $O(N \log N)$, and sliding-window $O(N)$ — were verified on 450 coprime pairs (a, b) with $b \leq 40$, using 10 random cone vectors each. All results agreed to tolerance 10^{-9} (tolerance 10^{-8} for $N > 500$, accommodating floating-point accumulation).

a	b	N	$O(N^2)$ ops	$O(N)$ ops	max error
3	5	4	16	16	5.3×10^{-15}
5	7	12	144	48	1.8×10^{-14}
7	11	31	961	124	1.1×10^{-13}
11	13	60	3600	240	2.8×10^{-13}
23	29	314	98596	1256	1.2×10^{-11}
37	41	721	519841	2884	4.3×10^{-11}
97	101	4801	23049601	19204	1.3×10^{-9}
All 450 pairs: PASS . Pairs with $N > 500$ verified against $O(N \log N)$ reference.					

TABLE 1. Two-generator verification.

Remark 7.1. The naive $O(N^2)$ implementation directly transcribes the definition of [3] and serves as ground truth for small N . For $(a, b) = (97, 101)$ its runtime becomes prohibitive, illustrating that the original formulation is too costly to benchmark its own optimisation at scale. The $O(N \log N)$ prefix-sum algorithm, itself verified against naive for small N , closes the verification chain transitively for large N .

Generators	N	Active windows	max error	Status
$\{2, 3\}$	1	2	0	OK
$\{3, 5\}$	4	2	8.9×10^{-16}	OK
$\{2, 3, 5\}$	1	4	0	OK
$\{3, 5, 7\}$	3	5	1.8×10^{-15}	OK
$\{2, 3, 5, 7\}$	1	8	0	OK

TABLE 2. Multi-generator verification against naive $O(N^2)$.

7.3. Multiple generators.

n	$2^n - 1$	$w(n)$	$M(n)$	Redund.%	Gap
2	3	2	0.3333	0.0%	-0.3333
3	7	4	0.2667	14.3%	-0.3048
4	15	8	0.2286	26.7%	-0.3048
5	31	14	0.2078	29.0%	-0.2438
6	63	20	0.1918	44.4%	-0.1257
7	127	30	0.1805	59.1%	-0.0557
8	255	38	0.1710	72.2%	+0.0220
9	511	48	0.1636	81.6%	+0.0697
10	1023	60	0.1579	88.0%	+0.0993
11	2047	84	0.1529	92.5%	+0.1118
12	4095	116	0.1487	95.3%	+0.1204
13	8191	146	0.1451	97.2%	+0.1273
14	16383	198	0.1417	98.3%	+0.1296
15	32767	236	0.1387	99.0%	+0.1315
16	65535	288	0.1361	99.4%	+0.1317
17	131071	352	0.1338	99.7%	+0.1311
18	262143	412	0.1316	99.8%	+0.1300
19	524287	492	0.1296	99.9%	+0.1287
20	1048575	564	0.1278	99.9%	+0.1273
21	2097151	646	0.1260	100.0%	+0.1257
22	4194303	720	0.1245	100.0%	+0.1243
23	8388607	808	0.1230	100.0%	+0.1229
24	16777215	892	0.1216	100.0%	+0.1215
25	33554431	996	0.1203	100.0%	+0.1203
26	67108863	1096	0.1191	100.0%	+0.1191
27	134217727	1200	0.1180	100.0%	+0.1180
28	268435455	1306	0.1169	100.0%	+0.1169
29	536870911	1416	0.1158	100.0%	+0.1158
30	1073741823	1528	0.1148	100.0%	+0.1148

TABLE 3. Active window count $w(n)$, Mertens product $M(n)$, redundancy rate, and $\text{Gap}(n) = M(n) - w(n)/(2^n - 1)$, generators $= p_1, \dots, p_n$. Three observed milestones: $n = 8$ (smallest n with $\text{Gap}(n) > 0$), $n = 21$ (smallest n with redundancy rate = 100%), $n = 25$ (smallest n with $\text{Gap}(n) = M(n)$ to four decimal places). Maximum observed Gap at $n = 16$ (bold). For $n \geq 25$, $\text{Gap}(n) = M(n)$ within the floating-point precision of our metric. Data for $n \leq 18$: direct enumeration. $n = 19, \dots, 30$: incremental algorithm (Section 5), verified independently at $n = 27, 28$ by direct $O(2^n)$ enumeration (Section 7.1).

7.4. Mertens comparison.

8. STRUCTURAL INTERPRETATION

8.1. Universality across generator sequences. The behaviour observed for the first n primes extends to arbitrary generator sequences. Table 4 summarises results for eight qualitatively different sequences at $n = 25$.

Sequence	First gen.	Gap > 0 at n	Gap/ M at $n = 25$	Gap/ M at $n = 300$
<i>Prime sequences</i>				
First n primes (baseline)	2	8	0.9998	1.0000
Primes from 101	101	2	0.9999	1.0000
Primes from 1009	1009	2	0.9997	1.0000
Every 2nd prime ($p_2, p_4, \dots$)	3	8	0.9998	1.0000
Every 3rd prime ($p_3, p_6, \dots$)	5	2	0.9998	1.0000
Lower twin primes	3	9	0.9996	1.0000
Primes $\equiv 3 \pmod{4}$	3	8	0.9998	1.0000
Primes $\equiv 1 \pmod{4}$	5	2	0.9998	1.0000
<i>Random sequences (GPT density test)</i>				
Random, same density (seed 42)	—	8	0.9999	—
Random, same density (seed 137)	—	8	0.9999	—
Random, no density match (seed 42)	—	2	0.9999	—

TABLE 4. Theorem 8.4 ($\text{Gap}/M \rightarrow 1$) holds for all tested sequences. At $n = 300$, $\text{Gap}/M(n) = 1.0000$ (six decimal places) for all prime sequences. Random odd integers with the same density as primes give identical Gap/M and crossover behaviour to the prime baseline — confirming the effect is a property of generator *density*, not arithmetic structure. Random integers without density matching behave like large-prime sequences (immediate crossover at $n = 2$).

Remark 8.1 (Density determines crossover). The crossover point ($\text{Gap} > 0$) and the rate of $\text{Gap}/M \rightarrow 1$ depend on the *density* of generators, not their arithmetic nature. Small generators (density: many generators per unit interval) delay the crossover; large generators (sparse density) produce immediate crossover at $n = 2$. Random odd integers with the same density as the prime sequence reproduce the prime baseline exactly ($n^* = 8$, $\text{Gap}/M = 0.9999$ at $n = 30$). This indicates that the mechanism is combinatorial rather than number-theoretic: the prime sequence is one example of a sequence with density $\sim 1/\log n$, and the empirical behaviour depends only on this density.

The efficiency gain has a transparent arithmetic explanation. The kernel $K^{(n)}$ is the inclusion–exclusion Möbius function over the generator set. Its sparsity reflects two independent effects.

- (i) *Mertens-type sparsity*. Each generator p_i eliminates a fraction $\sim 1/p_i$ of the gap set, reducing N . This accounts for the factor $M(n) = \prod(1 - 1/p_i)$.
- (ii) *Redundancy cancellation*. Distinct subsets with equal sums and opposite parities annihilate breakpoints in $K^{(n)}$, reducing $w(n)$ below what Mertens predicts. This effect grows with n (redundancy rate $0\% \rightarrow 99.8\%$ for $n = 2 \rightarrow 18$) and has no analogue in the prime sieve, where primes are by construction non-redundant.

The combination gives total complexity $O(w(n) \cdot N)$ decreasing faster than either factor alone.

8.2. Activation and deactivation of window positions. Each new generator p_{n+1} modifies $K^{(n+1)}$ relative to $K^{(n)}$ by adding the contribution of every subset containing p_{n+1} .

Lemma 8.2 (Activation mechanism). *Let $\mathbf{p}' = (p_1, \dots, p_n, p_{n+1})$. For any $d \in \mathbb{Z}$,*

$$K^{(n+1)}(d) - K^{(n)}(d) = \sum_{\substack{S \subseteq \{p_1, \dots, p_n\} \\ \sigma(S) + p_{n+1} \leq d}} (-1)^{|S|+1}.$$

In particular, a position d that is inactive under $K^{(n)}$ becomes active under $K^{(n+1)}$ if and only if the above sum is nonzero, i.e. the new generator introduces an imbalance between even- and odd-sized subsets whose augmented sum falls at or below d .

Proof. Direct from Definition 6.1: subsets not containing p_{n+1} contribute identically to $K^{(n+1)}$ and $K^{(n)}$. The difference is the contribution of subsets $S \cup \{p_{n+1}\}$, which equals $(-1)^{|S|+1} \cdot \mathbf{1}_{d \geq \sigma(S)+p_{n+1}}$. $\square$

Remark 8.3. The lemma shows that positions deactivated at step n can be reactivated at step $n+1$ if the new generator restores parity imbalance at that point. Empirically (Table 5), the first few generators are smallest primes and their pairwise sums are small, so reactivation is frequent for small positions and rare for large ones. Moreover, $K^{(n)}(d)$ is not restricted to $\{-1, 0, +1\}$: values ± 2 and ± 3 are observed at $n = 12$ (e.g. positions 47 and 149), indicating that multiple imbalanced subsets can accumulate at the same breakpoint. No nontrivial bound on $|K^{(n)}(d)|$ is currently known; this is a natural open question.

pos	$n=2$	$n=3$	$n=4$	$n=5$	$n=6$	$n=7$	$n=8$	$n=9$	$n=10$	$n=11$	$n=12$
0	1	1	1	1	1	1	1	1	1	1	1
3	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1
5	.	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1
7	.	.	-1	-1	-1	-1	-1	-1	-1	-1	-1
8	.	1	.	.	.	.	.	.	.	.	.
9	.	.	1	1	1	1	1	1	1	1	1
10	.	.	1	1	1	1	1	1	1	1	1
16	.	.	.	.	1	1	1	1	1	1	1
17	.	.	.	1	2	1	1	1	1	1	1
25	.	.	.	.	.	.	1	1	1	1	1
47	.	.	.	.	.	1	1	.	-1	-2	-3

TABLE 5. Selected active window positions and their $K^{(n)}$ values for $n = 2, \dots, 12$ (generators = $p_1, \dots, p_n$). A dot denotes $K^{(n)}(d) = 0$ (inactive). Position 8 activates at $n = 3$ and deactivates at $n = 4$. Position 17 reaches $K = 2$ at $n = 6$ (amplification). Position 47 reaches $K = -3$ at $n = 12$.

8.3. Density asymptotic theorem. Table 3 reveals that $\text{Gap}(n)$ is not monotone. It increases from $n = 8$ to a numerical maximum near $n = 16$ (≈ 0.1317), then decreases slowly. Empirically, by $n = 28$ the ratio $\text{Gap}(n)/M(n)$ reaches 1.0000 to four decimal places and remains there throughout the tested range $n = 2, \dots, 300$ across all eight sequence families of Section 7. The following theorem explains and formalises this behaviour.

Theorem 8.4 (Density asymptotic). *Let $\mathbf{p} = (p_1, p_2, \dots)$ be a sequence of distinct integers greater than 1 with $\gcd(p_1, \dots, p_n) = 1$ and $M(n) = \prod_{i=1}^n (1 - 1/p_i) \rightarrow 0$ as $n \rightarrow \infty$. Assume in addition the subexponential growth condition $\log p_n = o(n)$ (equivalently $p_n = e^{o(n)}$). Then*

$$\frac{w(n)}{2^n - 1} = o(M(n)) \quad \text{as } n \rightarrow \infty,$$

equivalently $\text{Gap}(n)/M(n) \rightarrow 1$.

The growth condition $\log p_n = o(n)$ is satisfied by every generator family considered in Section 7: the first n primes ($p_n \sim n \log n$), shifted primes, twin primes, arithmetic progressions of primes, and random odd integers with prime-like density.

Proof. We bound $w(n)/((2^n - 1)M(n))$ and show it tends to zero.

Step 1. $w(n) \leq \sigma_n := \sum_{k=1}^n p_k$ by Lemma 6.6. The bound is verified empirically to $n = 30$ (Table 3; see also `verify_v2.py`).

Step 2: bound on $H_n = \sum_{k=1}^n 1/p_k$. Since $p_1, \dots, p_n$ are distinct integers ≥ 2 , the harmonic-type sum H_n is maximised when $\{p_k\}$ is the set $\{2, 3, \dots, n+1\}$ of the n smallest such integers. Hence

$$H_n \leq \sum_{j=2}^{n+1} \frac{1}{j} \leq \log(n+1).$$

By $\log(1 - x) \geq -2x$ for $x \in [0, 1/2]$ and $1/p_k \leq 1/2$,

$$M(n) \geq e^{-2H_n} \geq (n+1)^{-2}.$$

Step 3: combining the bounds. Using $\sigma_n \leq n \cdot p_n$ (where $p_n = \max_k p_k$) and $2^n - 1 \geq 2^{n-1}$,

$$\frac{w(n)}{(2^n - 1) M(n)} \leq \frac{2\sigma_n}{2^n \cdot M(n)} \leq \frac{2np_n(n+1)^2}{2^n} = 2n(n+1)^2 \cdot \frac{p_n}{2^n}.$$

Step 4: subexponential growth implies convergence. Under the hypothesis $\log p_n = o(n)$,

$$\log\left(\frac{p_n}{2^n}\right) = \log p_n - n \log 2 = o(n) - n \log 2 \rightarrow -\infty,$$

so $p_n/2^n \rightarrow 0$ exponentially fast. Multiplying by the polynomial factor $2n(n+1)^2$ leaves the right-hand side tending to zero.

Therefore $w(n)/((2^n - 1) M(n)) \rightarrow 0$, equivalently $w(n)/(2^n - 1) = o(M(n))$, and $\text{Gap}(n)/M(n) = 1 - w(n)/((2^n - 1) M(n)) \rightarrow 1$. $\square$

Remark 8.5 (Density interpretation). The proof shows that the crossover point n^* (where $\text{Gap} > 0$) and the rate of convergence $\text{Gap}/M \rightarrow 1$ depend on the *harmonic density* $H_n = \sum 1/p_k$ of the generators: large generators contribute small $1/p_k$, so H_n grows slowly, redundancy saturates quickly, and n^* is small. Small generators contribute large $1/p_k$, requiring more steps before the exponential decay of 2^{-n} dominates. This is the formal expression of the threshold between *density-driven* and *count-driven* regimes.

The behaviour of $\text{Gap}(n)$ at finite n is consistent with diminishing increments: each new prime p_{n+1} contributes a fraction $\sim 1/p_{n+1}$ of additional redundancy relative to the existing structure, so increments to $\text{Gap}(n)$ decrease at a rate governed by the growth of p_n . By the prime number theorem $p_n \sim n \log n$, increments decay faster than $1/(n \log n)$, which is summable — consistent with the empirical convergence visible in Table 3. Whether $\text{Gap}(n)$ converges to a finite limit, reaches a true maximum at some finite n , or merely approaches its limit from above after a turning point, is not determined by the asymptotic statement and remains an open question (Section 10, item (iv)).

9. CONCLUSION

The present work makes four contributions.

(i) Linear-time evaluation of Q . The quadratic form Q of Huang [3] admits an $O(N)$ evaluation algorithm, reducing the naive $O(N^2)$ cost by a factor of N through direct arithmetic analysis of the interval structure of the kernel K .

(ii) Generalisation to n generators. The kernel $K^{(n)}$ defined by inclusion–exclusion over n generators has an interval structure that extends the two-generator case directly. This generalisation does not require solving the Frobenius problem for $n \geq 3$: the algorithm takes the gap set G as input and operates on its elements, independently of how G was computed. It is precisely Huang’s inclusion–exclusion formulation of K that makes this generalisation transparent — a structural property of his definition that was not the focus of the original work.

(iii) Incremental preprocessing. Active windows can be maintained incrementally as generators are added, reducing preprocessing from $O(2^n)$ to $O(n \cdot \sigma_n)$. This makes the algorithm practical for n up to several hundred.

(iv) Empirical exploration and asymptotic theorem. Numerical data to $n = 300$ (Section 7) for eight generator families (primes, shifted primes, twin primes, arithmetic progressions of primes, and random odd integers with prime-like density) exhibit three observed milestones ($n = 8$, $n = 21$, $n = 25$) and a parity-separated structure in the increments $\Delta w(n)$ for $n \geq 19$. Theorem 8.4 proves that $\text{Gap}(n)/M(n) \rightarrow 1$ under the subexponential growth condition $\log p_n = o(n)$, which is satisfied by all tested families. Across all tested sequences only the empirical crossover point n^* varies; the asymptotic behaviour is identical.

10. OPEN PROBLEMS

The following questions arise naturally from the numerical observations and the activation mechanism of Lemma 8.2. They are stated precisely to invite further investigation.

- (i) **Unboundedness of $K^{(n)}$.** Empirically, $|K^{(n)}(d)|$ takes values 2 and 3 for $n \leq 12$. Is $\sup_{n,d} |K^{(n)}(d)|$ finite or infinite? More specifically: for fixed d , does $K^{(n)}(d)$ stabilise as $n \rightarrow \infty$, or does it grow without bound?
- (ii) **Stable kernel of positions.** Empirically, a growing set of positions d appear to stabilise — their $K^{(n)}(d)$ value remains constant for all sufficiently large n . What is the mechanism that causes a position to stabilise? Is stabilisation permanent once it occurs, or can a position destabilise again for larger n ?
- (iii) **Crossover threshold n^* .** Theorem 8.4 (Section 8.3) proves $\text{Gap}(n)/M(n) \rightarrow 1$ but does not characterise the crossover point $n^* = \min\{n : \text{Gap}(n) > 0\}$. Empirically n^* depends on the harmonic density $H_n = \sum 1/p_k$: sequences with small generators (large $1/p_k$) have large n^* ; sequences with large generators have $n^* = 2$. Is there a closed-form expression for n^* in terms of H_n or the growth rate of p_k ?
- (iv) **Maximum of $\text{Gap}(n)$.** Empirically, $\text{Gap}(n)$ for the first n primes attains a numerical maximum near $n = 16$ (≈ 0.1317) and decreases monotonically thereafter (Table 3). Does $\text{Gap}(n)$ attain a true maximum at some finite n ? Equivalently: is the convergence $\text{Gap}(n)/M(n) \rightarrow 1$ from below for all sufficiently large n , or does the difference oscillate?
- (v) **Bipartite structure of $\Delta w(n)$.** From $n = 19$ onward, the increments $\Delta w(n) = w(n) - w(n-1)$ split into two linearly growing sequences according to the parity of n :

n	19	20	21	22	23	24	25	26
Δw	80	72	82	74	88	84	104	100
parity	odd	even	odd	even	odd	even	odd	even

Odd- n increments: 80, 82, 88, 104, 110, $\dots$ (increasing). Even- n increments: 72, 74, 84, 100, 106, $\dots$ (increasing). Both sequences grow approximately linearly, offset by ≈ 8 . This structure does not appear for $n < 19$ and coincides with the onset of redundancy saturation. What is the arithmetic mechanism behind this bipartite split?

- (vi) **Structure of position renewal.** Lemma 8.2 identifies when a new generator can reactivate a previously deactivated position. Is there a characterisation of which positions are renewed at each step, in terms of the arithmetic of the generators?
- (vii) **Asymptotic increment law $\Delta w(n) \sim p_n$.** Empirical data reveals a striking pattern at large n : the increment $\Delta w(n) = w(n) - w(n-1)$ converges to the generator value p_n itself.

For the baseline sequence (first n primes) at $n = 300$: $p_{300} = 1987$, $\Delta w(300) = 1986$ (difference: 1). For the sequence starting from 101 at $n = 300$: $p_{300} = 2153$, $\Delta w(300) = 2153$ (difference: 0).

The pattern $\Delta w(n) \approx p_n$ appears to hold for all tested sequences once n is sufficiently large (and exactly for sequences of large generators from the first step).

The proposed mechanism: when p_n is large relative to the existing subset sums, almost every subset $S \cup \{p_n\}$ produces a unique new breakpoint $\sigma(S) + p_n$ with no parity collision, contributing a net new active window. The number of such subsets is approximately 2^{n-1} but most cancel in pairs, leaving a residue of order p_n .

Formally: does $\Delta w(n)/p_n \rightarrow 1$ as $n \rightarrow \infty$ for sequences where $p_n \rightarrow \infty$? If so, this would give $w(n) \sim \sum_{k=1}^n p_k = \sigma_n$, providing an explicit asymptotic for $w(n)$ in terms of the generator sum.

Remark on related literature. The cancellation mechanism identified here appears independently in two other bodies of work, noted for context rather than as direct antecedents.

Whitney [6] observed that terms in inclusion–exclusion for graph chromatic polynomials cancel predictably; this was generalised by Chen and Qian [2] to ordering-free cancellations.

Those results concern graph-theoretic objects and do not address arithmetic kernels on numerical semigroups.

In analytic number theory, the exponential growth of inclusion–exclusion terms in sieve theory and the partial cancellations due to parity constraints are discussed by Tao [5] in the context of the parity problem. The connection to the present setting is structural: the Mertens-type sparsity of active windows and the redundancy cancellations we observe are arithmetically analogous to sieve parity phenomena, though the objects and proof methods differ.

We arrived at this connection by examining the arithmetic behaviour of $K^{(n)}$ directly, not by importing sieve-theoretic tools. The parallel emerged *a posteriori* during a search of related literature on inclusion–exclusion cancellations. To our knowledge, neither the interval-sparsity of $K^{(n)}$ nor the parity-cancellation mechanism for numerical semigroup kernels has been previously described.

Source code for all numerical verifications is available as `verify_linear_Q.py`, `appendix_verify.py`, and `verify_v2.py`.

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